

ZcannerReality: An AI and Augmented Reality-Based Universal Virtual Try-On Platform

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Abstract

The rapid growth of e-commerce has increased the demand for online fashion shopping, but customers often cannot visualize how products such as clothing, shoes, watches, sunglasses, and other accessories will look before purchasing, leading to poor buying decisions and increased product returns. Existing virtual try-on solutions are mostly limited to specific brands and do not support multiple fashion categories on a single platform. To address this challenge, this project proposes ZcannerReality, an Artificial Intelligence (AI) and Augmented Reality (AR) based universal virtual try-on platform. The system uses deep learning to identify products, computer vision to detect the corresponding body part, and augmented reality to overlay the selected product in real time, providing a realistic and interactive virtual try-on experience. The proposed platform aims to improve customer confidence, reduce product returns, and enhance the online shopping experience through a user-friendly and scalable solution. By integrating AI, Computer Vision, and AR into a single platform, ZcannerReality offers a practical and immersive approach to digital fashion, helping users make better purchasing decisions before buying products online.

Keywords: **Artificial Intelligence; Deep Learning; Augmented Reality; Computer Vision; Virtual Try-On; E-Commerce; Fashion Technology.**